

Automated Traffic Violation Detection System

System Design Overview

Introduction

The system described in this document is an end-to-end pipeline that converts raw traffic-camera footage into structured, evidence-backed violation records suitable for transmission to a concerned authority. The pipeline is organized as a sequence of cooperating stages: image preprocessing, vehicle detection and tracking, a set of per-vehicle violation classifiers (helmet, seatbelt, triple-riding, speed, stop-line, red-light, illegal parking, and wrong-side driving), a temporary evidence database keyed by vehicle identity, a conditional second preprocessing pass restricted to flagged vehicles, and a license-plate recognition stage whose output is merged with the violation record before being emailed to the relevant authority or written to a permanent store. A separate, network-level layer correlates evidence for a flagged vehicle across multiple cameras. Each stage below is described at a level sufficient to understand its role and its key design decisions; the accompanying technical documentation files listed for each stage contain the full mathematical formulation, configuration parameters, and reference implementation.

1 Image Acquisition and Preprocessing

The pipeline begins with a raw frame captured by a CCTV camera at an intersection, with no assumption made about illumination, weather, or motion-blur conditions. Rather than applying a fixed, always-on enhancement filter, the system first computes a **cumulative darkness score** from four normalized image statistics — histogram spread, percentage of dark pixels, mean brightness, and entropy — each weighted by an empirically tuned coefficient. Only frames whose score meets or exceeds a calibrated threshold are routed through **Zero-DCE++**, a lightweight, unsupervised low-light enhancement network chosen for its robustness across arbitrary lighting and its suitability for near-real-time inference at the edge. Frames that pass through enhancement are subsequently denoised with a **bilateral filter**, which is applied *after* enhancement specifically because curve-based brightening amplifies sensor noise that was previously masked in dark regions. Both the enhanced and the already-well-lit paths converge at a single, unconditional deblurring step using **Wiener deconvolution**, since motion blur from moving vehicles is a function of vehicle and camera dynamics rather than ambient light, and is therefore corrected on every frame regardless of the darkness decision.

A second, independent sub-pipeline is triggered only after a violation has already been flagged. Once a vehicle's bounding box localizes its number-plate region, the cropped region is passed through **LPDGAN**, a conditional generative adversarial network trained with a combined adversarial and pixel-reconstruction (L1) loss, that performs a second, targeted restoration pass tolerating residual blur or low light that the frame-level correction could not fully resolve. This keeps the computationally expensive GAN-based restoration off the real-time critical path, since it is invoked only for the comparatively rare subset of vehicles already classified as violators (*see `image_preprocessing_pipeline.pdf` for full detail*).

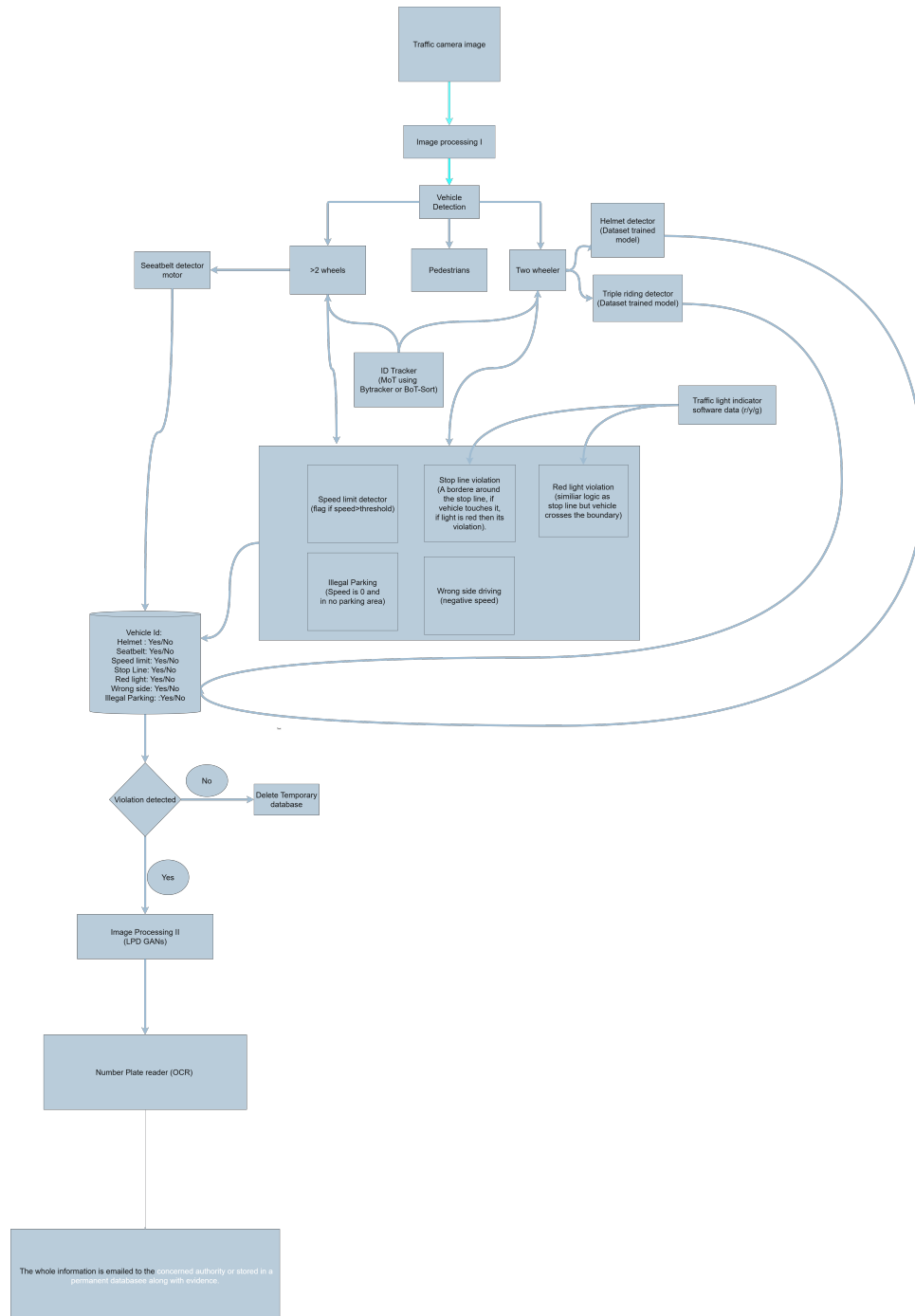


Figure 1: System Architecture.

2 Vehicle Detection and Multi-Object Tracking

Object localization is performed by **YOLO26**, queried once per frame to return, for every detected object, a bounding box, a COCO class label, and a confidence score; the nano-scale checkpoint is used for edge-constrained deployment. Detections are linked across frames using **BoT-SORT**, which extends the motion-and-IoU-based ByteTrack with two additional mechanisms suited to a non-static camera and visually similar nearby objects: **Global Motion Compensation**, which estimates the camera’s own inter-frame motion from sparse keypoints and warps each track’s predicted state before matching, preventing camera shake from being misread as object motion; and an optional **Re-Identification** embedding, fused into the cost matrix to reduce identity switches between similar, closely-passing objects. Frame-to-track assignment is solved as a linear bipartite matching problem (the Hungarian algorithm) over a cost combining spatial and, where enabled, appearance distance, with unmatched detections seeding new tracks and unmatched tracks retained as lost for a configurable buffer before termination.

Detected vehicles are split downstream into three categories — pedestrians, two-wheelers, and vehicles with more than two wheels — each routed to the violation logic relevant to that class. Rather than passing every frame of a vehicle’s track to the downstream classifiers, a **crop-extraction** stage reduces each track to exactly one representative image: occupants are associated to a vehicle using **Intersection over the person box’s own Area (IoA)** rather than IoU, since the question being asked is what fraction of a person’s body overlaps the vehicle rather than how similarly sized the two boxes are; for cars, association is restricted to an upper-body zone of the vehicle box, since the seatbelt signal is confined to that region in any case. Across a track’s lifetime, the single best frame is retained using rider count as the primary key and the area of the smallest enclosing box (a clarity proxy) only as a tie-breaker, since under-counting riders could cause a genuine triple-riding violation to be missed. The track is emitted once the tracker reports no detection for it beyond a fixed timeout, guaranteeing exactly one crop per physical vehicle appearance (*see [multi_vehicle_seatbelt_helmet_triple_detector.pdf](#) for full detail*).

3 Per-Vehicle Compliance Classifiers

Three classifiers operate on the crops emitted above. The **helmet detector** and **seatbelt detector** are dataset-trained models evaluating, respectively, two-wheeler riders and car occupants for the presence of a helmet or an engaged seatbelt strap, trained on purpose-built public datasets. The **triple-riding detector** requires no separate model at all: because the crop-extraction stage already associates occupant boxes to a motorcycle via the IoA criterion, a triple-riding violation is deterministically flagged whenever the retained occupant count for the best frame is three or more, avoiding the overhead of an additional dense-prediction model (*see [multi_vehicle_seatbelt_helmet_triple_detector.pdf](#) for full detail*).

4 Speed Estimation

Vehicle speed is computed from the same tracked bounding boxes via a five-stage pipeline: extracting each vehicle’s ground-contact reference point (the bottom-center of its box, the closest 2D approximation to its true road-contact location); projecting that pixel coordinate into real-world metric coordinates via a **homography matrix** calibrated once per camera from four known point correspondences; **plausibility gating**, which rejects any displacement implying a physically impossible speed and clears the affected track’s history entirely rather than discarding only the offending point, since a corrupted track should not silently resume averaging after a discontinuity; and finally computing the windowed Euclidean displacement over a sliding window of history, using elapsed time derived from the video’s native frame rate, and smoothing the

result with an **Exponential Moving Average** to suppress the frame-to-frame jitter that persists even at constant true speed due to bounding-box noise at the window’s endpoints. The **speed limit detector** simply flags a vehicle whose smoothed speed exceeds a configured threshold (*see [speed_calculation_4_final.pdf](#) for full detail*).

5 Stop-Line and Red-Light Violation Detection

This stage divides the frame into a Normal Zone before the stop line and a Detection Zone beyond it, using a half-plane test against the stop-line geometry. Critically, the current **traffic signal color is not inferred from the image** but is supplied directly by the traffic control system, which eliminates errors from glare, occlusion, adverse weather, or poor camera angle as a source of signal-misreading. If the signal is not red, every vehicle is immediately classified as no anomaly. If the signal is red and a vehicle’s centroid lies in the Detection Zone, an anomaly is confirmed, and the vehicle’s already-computed speed disambiguates the violation type: a stationary vehicle past the line is a **stop-line violation**, while a still-moving vehicle past the line is an active **red-light violation** (*see [red_light_and_stop_line_violation.pdf](#) for full detail*).

6 Illegal Parking and Wrong-Side Driving Detection

This module branches on a single test: whether a tracked vehicle’s speed is zero. A stationary vehicle is checked against pre-segmented **no-parking zones**; a per-vehicle dwell timer accumulates the time spent at zero speed inside such a zone, and the vehicle is flagged as **illegally parked** only once this timer exceeds a *class-dependent* grace period (nominally 10 seconds for cars versus 60 seconds for buses), which absorbs brief, legitimate halts such as passenger drop-off. A moving vehicle is instead checked against a pre-defined directional sign convention for that road segment; a mismatch between the sign of its lateral position-times-speed product and the zone’s defined convention indicates **wrong-side driving**. Both checks reuse the same enhanced frame and zone segmentation already computed upstream, requiring no additional model inference (*see [illegal_parking_wrong_side_driving.pdf](#) for full detail*).

7 Temporary Evidence Database and Violation Gate

Every per-vehicle outcome — vehicle ID, and a Yes/No flag for each of helmet, seatbelt, speed limit, stop line, red light, wrong side, and illegal parking — is written to a temporary, per-vehicle record. A single gate, “*Violation detected*,” then checks whether any flag in that record is positive. If none are, the temporary record is deleted immediately, ensuring that compliant vehicles leave no trace in the system and that storage grows only with genuine violations. If at least one flag is positive, the vehicle proceeds to the final stages below.

8 Conditional Second-Pass Imaging and Plate Recognition

Only vehicles that clear the violation gate are passed through **Image Processing II**, which invokes the LPDGAN-based plate restoration sub-pipeline described in Section 1, cropping and repairing the number-plate region before it reaches **Optical Character Recognition (OCR)**. The OCR output is accepted only above a defined confidence threshold and with the expected character length for a standard plate format, after which it is merged back into the vehicle’s violation record (*see [image_preprocessing_pipeline.pdf](#) for full detail*).

9 Cross-Camera Evidence Correlation

A network-level layer sits above the single-camera pipeline described in the preceding sections. Each camera continuously writes its local findings — including vehicles with no violations — into a **shared state** store; an entry is deleted if it carries no violation flag, and retained otherwise. The physical camera network is modeled as a **topology graph**, with edge weights representing a learned mean and variance of historical transit time between camera pairs (rather than a naive distance-over-speed-limit estimate), since real transit time depends on the number of intersections and signals along a route as much as on raw distance. For a flagged vehicle, this graph is used to compute, for each reachable downstream camera, a **time search space** — an arrival window centered on the historical mean and widened by its standard deviation — rather than a single predicted timestamp, since a vehicle may be delayed by a signal or, if originally flagged for speeding, may arrive earlier than a naive estimate would suggest. When a downstream camera’s clock falls within this window, the system attempts plate-based confirmation via OCR, falling back to appearance-based matching when OCR confidence or extracted plate length is insufficient, before merging the new sighting into the vehicle’s case file (*see [evidence_extractor_2.pdf](#) for full detail*).

10 Final Reporting

Once a vehicle’s record contains a confirmed violation, an extracted plate number, and any cross-camera evidence, the complete record — together with supporting image evidence — is emailed to the concerned authority or written to a permanent database, closing out the pipeline for that vehicle.

Each section above summarizes the corresponding stage at a high level. The referenced technical documentation files contain the full mathematical formulation, tunable parameters, and reference code for that stage and should be consulted for implementation detail.